ADNI · ADNIMERGE2 V0.1.2 · DATA CUT-OFF 2026-07-06

APOE ε4 and the rate of cognitive decline: a longitudinal mixed-model analysis of ADAS-Cog 13

How fast does cognition worsen over six years of follow-up, and does the number of APOE ε4 alleles change that rate once baseline age, sex, education and diagnosis are accounted for? This report also looks at whether missed visits bias the answer.

2,830 participants · 8,782 visits Analysis: SAS 9.4 (SAS OnDemand) · Stata replication pending

Independent check: R nlme

Key findings

1. **Decline speeds up with each ε4 allele.** ADAS-Cog 13 rises (worsens) by **1.18** points/year with no ε4, **1.98** with one and **2.86** with two (APOE4 × time $F(2, 5945) = 43.1$, $p < 0.001$). Each allele adds about 0.8 points/year.
2. **The APOE4 effect is concentrated in MCI and dementia.** Carriers decline 1.3 points/year faster than non-carriers in MCI and 1.2 faster in dementia, but only 0.35 faster in cognitively normal participants (three-way interaction $p = 0.0001$).
3. **People differ a lot in how fast they decline.** The person-to-person SD of the annual slope is 2.0 points, larger than the average ε4 effect. People who start with worse scores also decline faster (intercept–slope correlation 0.60).
4. **Missing data are not MCAR but look compatible with MAR.** A worse score at the last visit predicts missing the next one (OR 1.046 per point, LR $\chi^2 = 96.6$). The ε4 slope differences are nearly identical under a pattern-mixture model, and in the complete-case analysis the absolute slopes fall but the ε4 differences stay.
5. **The finding holds for a second outcome.** In a binomial GLMM for MMSE errors, the odds of an item error grow 6% per year with no ε4, 15% with one and 25% with two.

> Results below come from the SAS run. Every model converged and the log has no errors. An independent R (`nlme`) refit reproduces the SAS fixed effects, variance

errors. An independent R (nlme) refit reproduces the SAS fixed effects, variance components and slopes to every printed digit. The Stata do-file will be run as a second replication once a licence is available.

1 Data and analysis sample

Data come from the ADNIMERGE2 R package. Subject-level covariates (age, sex, education, baseline diagnosis, APOE genotype) are taken from `ADSL`, and repeated cognitive scores from `ADQS`. ADNI protocols schedule visits differently: ADNI1 visits fall at months 6, 12, 18 and 24, while ADNI2–4 visits are yearly. Each visit was therefore mapped to the nearest anniversary within ±0.35 years, giving nominal years 0–6. Semi-annual and off-cycle visits fall outside that window. Models use the *actual* time of each visit, not the nominal year.

Of 2,974 enrolled participants, 2,830 had complete baseline age, sex, education, diagnosis, APOE genotype and a baseline ADAS-Cog 13, and form the analysis sample.

Baseline characteristic	0 ε4	1 ε4	2 ε4
Participants, n	1,551	1,013	266
Age, mean (SD), years	72.7 (7.7)	72.5 (7.1)	70.2 (7.4)
Education, mean (SD), years	16.2 (2.7)	16.0 (2.8)	16.0 (2.6)
ADAS-Cog 13, mean (SD)	13.4 (8.3)	17.4 (10.1)	20.2 (9.6)
Baseline diagnosis, n (row %)			
Cognitively normal	759 (67%)	323 (29%)	45 (4%)
MCI	645 (51%)	481 (38%)	131 (10%)
Dementia	147 (33%)	209 (47%)	90 (20%)
Other			
Female, n (%)			
Protocol at enrolment: ADNI1 / GO / 2 / 3 / 4			810 / 128 / 77

ε4 dose is strongly confounded with baseline diagnosis (20% of the dementia group carry two copies, against 4% of the cognitively normal group). This is why every model adjusts for diagnosis and lets it modify the slope.

2 Methods

Restructuring

The R extraction writes one row per participant, with the outcome and visit time at years 0–6 in separate columns (wide format). SAS converts this to one row per participant-year (long format) with a DATA step: an array loop over `year = 0 to 6` that executes `output` on each pass. Stata does the same conversion with `reshape long`. Rows with a missing outcome are kept in the long file so missingness itself can be modelled.

Linear mixed-effects model (PROC MIXED, REML)

```
ADAS13_ij = β0 + β1·t_ij + β2·APOE4_i + β3·age_i + β4·dx_i + β5·sex_i + β6·educ_i
          + t_ij·(β7·APOE4_i + β8·age_i + β9·dx_i + β10·sex_i)
          + u0_i + u1_i·t_ij + ε_ij

(u0_i, u1_i) ~ N(0, G),  G unstructured;   ε_ij ~ N(0, σ²)
t = years since baseline;  age centred at 72;  education centred at 16 y;  APOE4 = 0,
```

Slopes by ε4 dose are `ESTIMATE` statements whose weights for the other covariates equal their observed means in the estimation sample. This matches Stata's `margins apoe4, dydx(time)`. Degrees of freedom use the between-within method. Each participant's predicted annual change was computed from conditional predictions at $t = 0$ and 1. That change is the fixed slope implied by their covariates plus their random-slope BLUP.

Missing data

- **Administrative censoring was separated from dropout.** A visit counts as missing only if its window closed before the data cut-off. Recently enrolled ADNI4 participants are not counted as dropouts.
- **MCAR vs MAR.** A discrete-time logistic model was fitted for missing visit k among those seen at k–1. Model A has baseline covariates only. Model B adds the most recent observed score. The models are compared by LR test and refitted as a GLIMMIX model with a random intercept.
- **MNAR sensitivity.** A complete-case model includes only participants retained to their last eligible visit. A pattern-mixture model lets intercept and slope differ by dropout pattern; its marginal slopes average over the observed mix of patterns.

Generalized linear mixed model (PROC GLIMMIX)

MMSE has a strong ceiling in cognitively normal participants. It was modelled as the number of item errors (30 – MMSE) out of 30, binomial with a logit link, with random intercept and slope, using Laplace estimation.

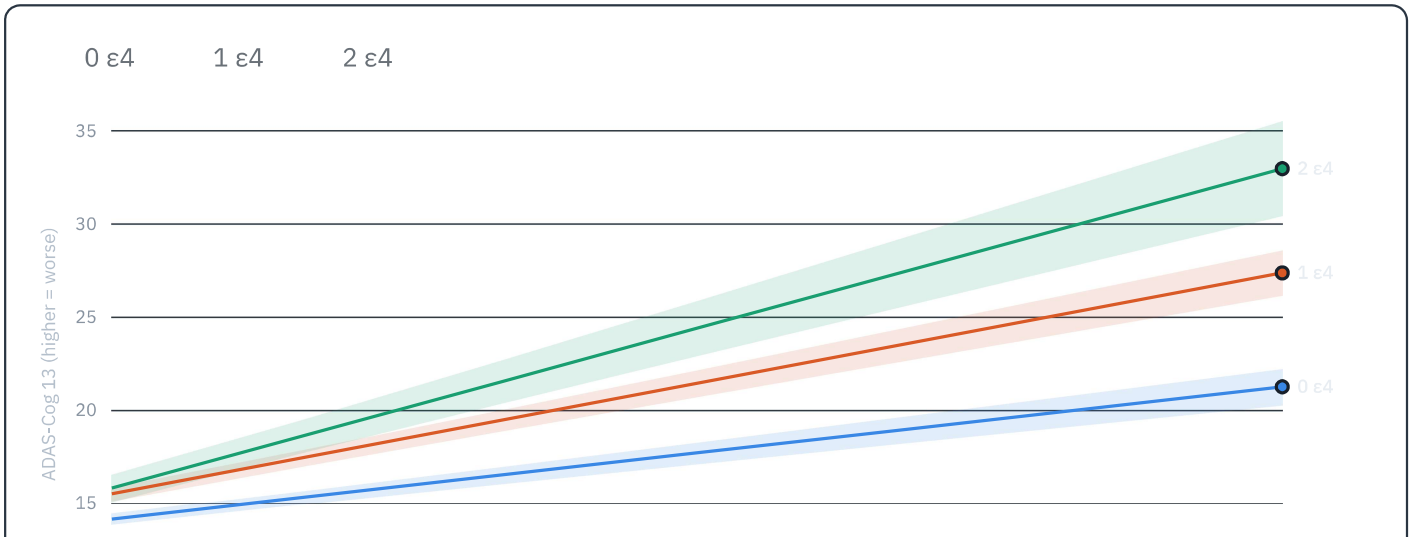
3 Choosing the random-effects structure

Unconditional growth model (ML)	−2 log L	Comparison	LR χ^2
Random intercept	60,238.3		
+ random slope, independent	58,101.9	vs random intercept (χ^2_0:χ^2_1 mixture)	2,136.4
+ intercept–slope covariance (unstructured)	57,355.7	vs independent (1 df)	746.2

Both the random slope and the intercept–slope covariance improve fit by a wide margin, so all later models use random intercepts and slopes with an unstructured G matrix.

Adding a fixed quadratic time term to the adjusted model also improved fit (LR $\chi^2 = 129.6$, 1 df, $p < 0.001$; coefficient +0.16 points/yr², from the R cross-check). Decline therefore accelerates. The linear slopes in this report are average rates over the follow-up window, which is how they are interpreted below.

4 Rate of decline by APOE ε4 dose





Model-predicted ADAS-Cog 13 with 95% confidence bands, averaged over the sample's age, sex, education and diagnosis mix. Higher scores mean worse cognition. The lines are straight because the model is linear in time; the quadratic check in section 3 shows real trajectories curve upward.

Annual change in ADAS-Cog 13	Estimate	SE	95% CI	p
0 ε4 alleles	1.18	0.074	1.04 to 1.33	< 0.001
1 ε4 allele	1.98	0.092	1.80 to 2.16	< 0.001
2 ε4 alleles	2.86	0.197	2.48 to 3.25	< 0.001
Differences in annual change				
1 vs 0	0.80	0.119	0.56 to 1.03	< 0.001
2 vs 0	1.68	0.213	1.26 to 2.09	< 0.001
2 vs 1	0.88	0.216	0.46 to 1.31	< 0.001

Global test of APOE4 × time: $F(2, 5945) = 43.13$, $p < 0.001$. The two allele steps are similar (0.80 and 0.88), consistent with an additive dose effect.

Full fixed-effects table

Term	Estimate	SE	p
Baseline level (ADAS-Cog 13 points)			
Intercept (CN, male, 0 ε4, age 72, 16 y education)	9.09	0.24	< 0.001
1 ε4 vs 0	1.35	0.25	< 0.001
2 ε4 vs 0	1.67	0.41	< 0.001
Age, per year	0.155	0.016	< 0.001
MCI vs CN	6.46	0.25	< 0.001
Dementia vs CN	19.65	0.36	< 0.001
Female vs male	−1.23	0.23	< 0.001
Education, per year	−0.282	0.042	< 0.001
Annual change (points/year)			

Time (CN, male, 0 ε4, age 72)	−0.04	0.11	0.73
× 1 ε4	0.80	0.12	< 0.001
× 2 ε4	1.68	0.21	< 0.001
× age, per year	0.022	0.008	0.006
× MCI	1.25	0.12	< 0.001
× Dementia	4.12	0.21	< 0.001
× female	0.21	0.11	0.054
Variance components (REML)			
SD random intercept	5.27	variance 27.79 (SE 0.97)	
SD random slope	2.01	variance 4.04 (SE 0.22)	
Correlation intercept–slope	0.60	covariance 6.35 (SE 0.34)	
SD residual	3.36	variance 11.27 (SE 0.25)	

Reference levels are 0 ε4, CN and male. SAS's GLM coding uses the last level as reference (2 ε4, dementia, female), so its printed SolutionF table looks different, but it is the same model with identical fit. A cognitively normal male non-carrier aged 72 shows no measurable decline (−0.04/yr), consistent with practice effects offsetting early change. Each extra year of baseline age adds 0.02 points/year to the rate of decline.

Does the ε4 effect depend on diagnosis?

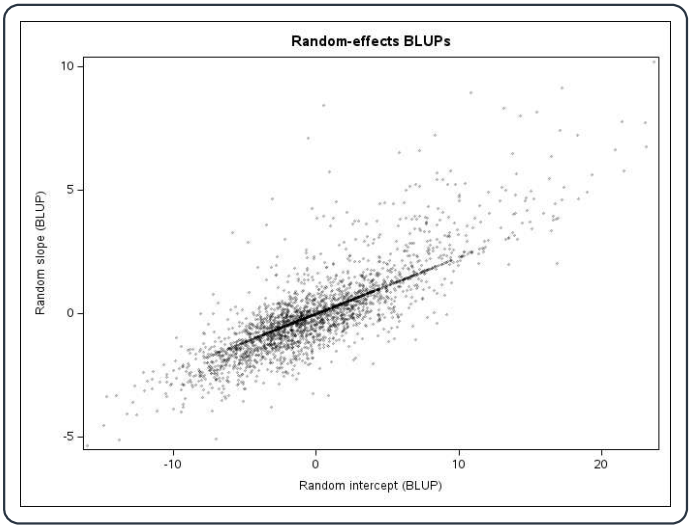
Carrier (≥1 ε4) minus non-carrier, annual change	Estimate	SE
Cognitively normal	0.35	0.18
MCI	1.34	0.15
Dementia	1.22	0.40

SAS Type 3 test of carrier × diagnosis × time: F(2, 5944) = 8.83, p = 0.0001. The group-specific estimates come from the R cross-check of the same model. The updated SAS script now prints them as ESTIMATE statements. The ε4 effect in cognitively normal people is small and borderline, which fits ε4 acting mainly once the disease process is under way.

5 Individual trajectories

Each participant's predicted annual change combines the fixed slope for their covariates with their random-slope BLUP. These summaries are not adjusted for the diagnosis mix, so carrier groups look further apart than the adjusted slopes in section 4.

Predicted individual annual change	n	Mean	SD	Q1	Median	Q3
By APOE ε4 dose						
0 ε4	1,551	0.99	1.83	−0.14	0.44	1.44
1 ε4	1,013	2.32	2.40	0.47	1.69	3.68
2 ε4	266	3.71	2.40	1.93	3.50	5.04
By baseline diagnosis						
Cognitively normal	1,127	0.37	0.80	−0.18	0.24	0.77
MCI	1,257	1.79	1.99	0.27	1.40	2.94
Dementia	446	4.95	2.28	3.45	4.67	6.24



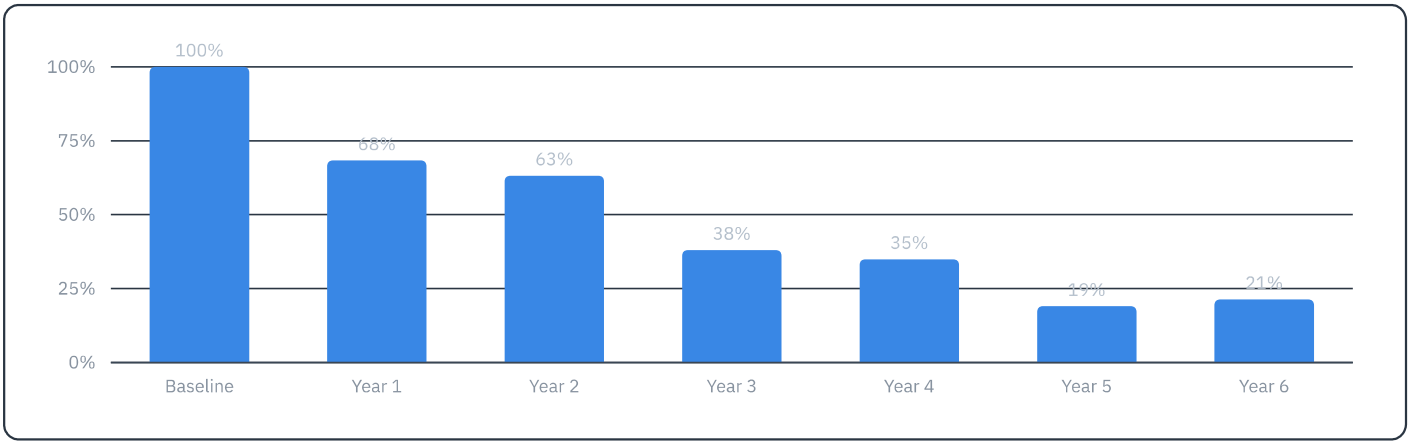
Random-effect BLUPs (from SAS). The positive tilt is the 0.60 intercept–slope correlation: people who start worse decline faster. The dense diagonal streak comes from participants with only a baseline visit, whose BLUPs are shrunk along the G-matrix correlation.

Shrinkage

For the 1,584 participants with three or more visits, BLUP-based slopes correlate 0.91 with slopes from a separate regression per person. Their SD is smaller (2.28 vs 3.01). The mixed model pulls noisy individual slopes toward the covariate-specific mean, which is the point of using BLUPs for prediction.

6 Missing data

How much is missing



Share of administratively eligible participants with an ADAS-Cog 13 at each nominal year. Eligible n falls from 2,830 to 2,219 as recent enrolees run out of time. Note that 685 participants (24%) have only a baseline visit, and only 172 (6%) are observed at all seven yearly visits.

There are 60 missingness patterns, and many are intermittent rather than monotone. For example, 82 people were seen at years 0, 1, 2 and 4 but missed year 3. Some of these gaps reflect protocol visit schedules and the ±0.35-year window, not participant choice.

Who misses visits

Dropped out before last eligible visit	Dropout	Test
Cognitively normal / MCI / Dementia	63.6% / 69.1% / 90.8%	$\chi^2(2) = 114.9$, $p < 0.001$
0 / 1 / 2 ε4 alleles	68.3% / 72.0% / 75.9%	$\chi^2(2) = 8.2$, $p = 0.016$
Baseline ADAS-Cog 13, dropouts vs retained	16.6 vs 12.9	$t = 10.7$, $p < 0.001$
Baseline age, dropouts vs retained	72.7 vs 71.6	$t = 3.9$, $p < 0.001$
Deaths among participants	268 (9.5%), similar across dropout patterns (8–11%)	

Dropout depends on observed baseline characteristics, so the data are not MCAR. In a multivariable model, diagnosis, baseline score and enrolment protocol were the significant predictors (c-statistic 0.75). APOE4 dropped out once diagnosis was included ($p = 0.81$).

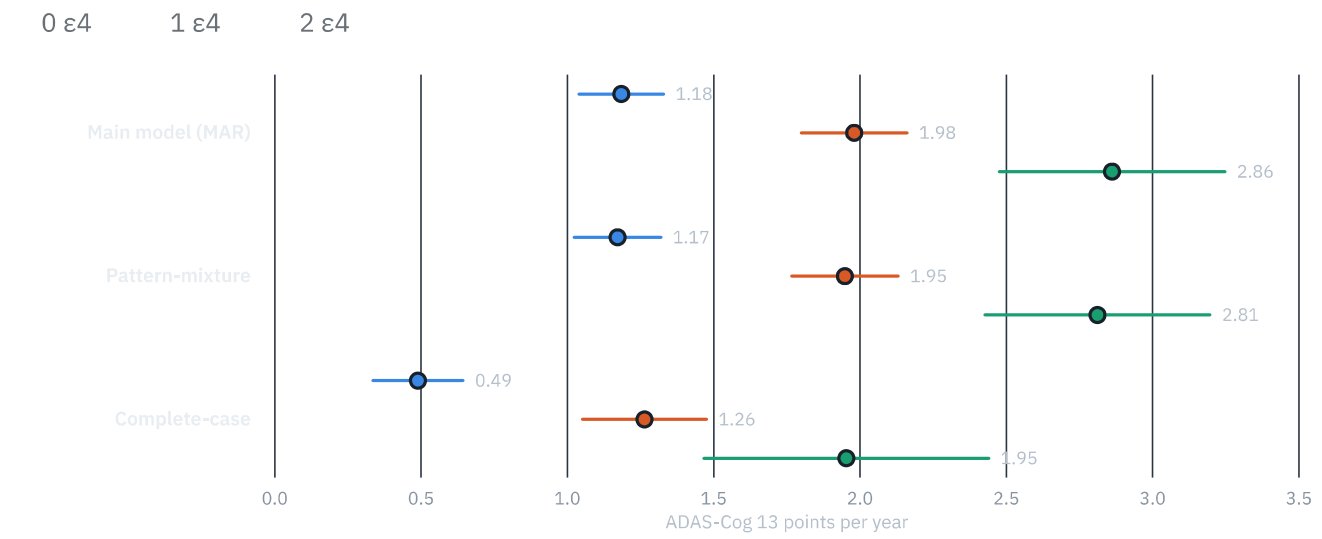
MCAR vs MAR: does the last observed score predict the next missed visit?

Next-visit missingness, 7,943 at-risk visits	OR	95% CI	p
Model B: most recent observed ADAS-Cog 13, per point	1.046	1.037 to 1.056	< 0.001
Baseline ADAS-Cog 13, per point (given most recent)	0.976	0.963 to 0.989	< 0.001
MCI vs CN	0.29	0.26 to 0.34	< 0.001

Age, per year	0.987	0.980 to 0.995	0.001
GLIMMIX with random intercept: most recent score, per point	1.051	1.040 to 1.061	< 0.001

LR test of model A (baseline covariates) vs model B (+ most recent score): $\chi^2(1) = 96.6$, $p < 0.001$. Missingness tracks the evolving observed trajectory, not just baseline. People whose scores have recently worsened are more likely to miss the next visit. This is the definition of MAR: missingness depends on observed data. Likelihood-based mixed models stay unbiased under MAR without modelling the dropout. The GLIMMIX random-intercept variance (0.34) shows some people are persistently more likely to miss visits, and the per-point effect is unchanged. Protocol and year carried large effects, consistent with visit schedules that differ across ADNI phases.

Sensitivity to missing-not-at-random



Annual change by ε4 dose (points and 95% CIs) under three missing-data assumptions.

Analysis	Participants	0 ε4	1 ε4	2 ε4	1 vs 0	2 vs 0
Main model (MAR)	2,830	1.18	1.98	2.86	0.80	1.68
Pattern-mixture	2,830	1.17	1.95	2.81	0.78	1.64
Complete-case (retained only)	839	0.49	1.26	1.95	0.77	1.46

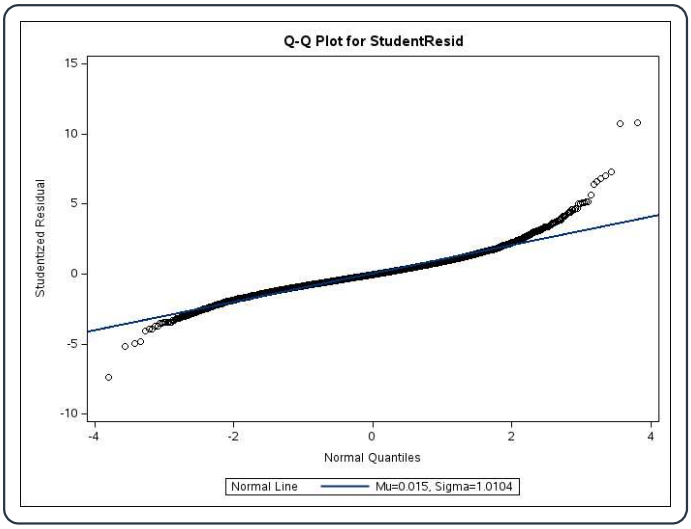
In the pattern-mixture model, participants who dropped out by year 2 declined 0.62 points/year faster than retained participants (95% CI 0.27 to 0.97, $p < 0.001$). Later dropouts declined 0.21 faster ($p = 0.084$). Overall test of slope differences by pattern: $F(2, 5943) = 6.10$, $p = 0.002$. Allowing for that difference barely moves the ε4 estimates. The complete-case analysis keeps only the 30% of participants who stayed in follow-up, who were healthier at baseline (mean ADAS-Cog 13 of 12.9 vs 16.6). Every absolute slope falls by 0.7–0.9 points/year, but the between-genotype gaps stay almost the same. The APOE4 conclusion is robust to these departures from MAR.

7 Second outcome: MMSE errors (binomial GLMM)

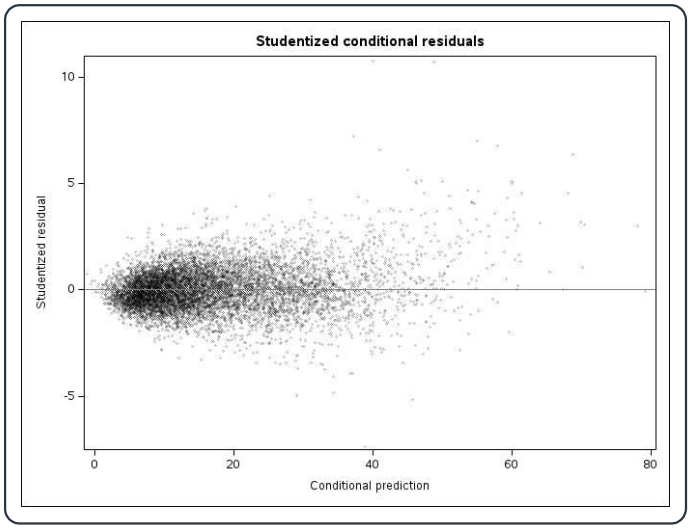
Odds of an MMSE item error, per year	OR	95% CI
0 ε4	1.060	1.037 to 1.084
1 ε4	1.152	1.123 to 1.182
2 ε4	1.252	1.190 to 1.316
Ratio of yearly ORs		
1 vs 0 ε4	1.087	1.051 to 1.124
2 vs 0 ε4	1.181	1.117 to 1.248

2,829 participants and 8,869 visits, Laplace estimation, random intercept and slope (unstructured). The test of the random-slope variance was $\chi^2 = 1{,}091.6$, $p < 0.001$ (mixture p-value). Conditional Pearson $\chi^2/df = 0.68$, so there is no over-dispersion. Modelling MMSE on its natural 0–30 error scale reproduces the ε4 dose–response seen with ADAS-Cog 13.

8 Model diagnostics



Studentized conditional residuals follow the normal line through the centre (SD 1.01), but the upper tail is heavy: the 99th percentile is 3.0 against 2.4 expected.



Residual spread grows with the predicted score. Variability is larger in the impaired range, where ADAS-Cog 13 is not bounded by a floor.

If the mean model is correct, fixed-effect estimates stay consistent under these departures, but model-based standard errors may be somewhat optimistic for the most impaired participants. Natural next checks are a residual variance that differs by diagnosis (`REPEATED / GROUP=dx` in PROC MIXED) or sandwich standard errors (`EMPIRICAL`). With

χ^2 statistics this large, neither is likely to change any conclusion.

9 Limitations

- **Linear time is a simplification.** Decline accelerates (section 3), so slopes describe average change over up to six years of follow-up. Groups followed for different lengths of time are averaged over different parts of the curve.
- **Follow-up is shortest for the people who decline fastest.** 91% of the dementia group dropped out before their last eligible visit. Dementia slopes rest mostly on the first two years.
- **Death truncates follow-up.** 9.5% of participants died. Their unobserved scores are a form of MNAR that the pattern-mixture model only partly addresses.
- **Annual mapping drops some visits.** The ±0.35-year window removes semi-annual visits (months 6 and 18) and off-cycle visits, which lowers precision.
- **ADNI is a volunteer research cohort, not a population sample.** Participants are highly educated (mean 16 years), which limits generalisability.

10 Reproducibility

Step	File	
Extract ADSL + ADQS, build the wide file	`extract_adni.R`	
SAS: DATA-step reshape, PROC MIXED, PROC GLIMMIX	`adni_longitudinal.sas`	Run on SAS OnDeman
Stata: reshape long, mixed, margins, predict	`adni_longitudinal.do`	
Independent check of key estimates	`R nlme`	

Since this run, the SAS script has been updated to print the APOE4 effect within each diagnosis and the sign of the quadratic term, and to suppress two cosmetic warnings. Re-running it will put those numbers in the SAS output directly.

Data used in this report were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). This page contains aggregate results only; no participant-level data are shown.